

BANK ACCOUNT SYSTEM

Objective / Aim

To design and implement a Bank Account System in Java demonstrating attributes, methods, constructors, object creation, inheritance, method overriding, deposits, withdrawals, interest calculation, and fund transfers.

Algorithm / Logic

1. Create the Account base class with account number, holder name, balance, and account type.
2. Define the constructor and deposit(), withdraw(), transfer(), and displayAccountDetails() methods.
3. Create SavingsAccount extending Account and add interestRate.
4. Define calculateInterest() in SavingsAccount.
5. Create CurrentAccount extending Account and add overdraftLimit.
6. Override withdraw() in CurrentAccount to permit withdrawals up to the overdraft limit.
7. Create SavingsAccount and CurrentAccount objects.
8. Perform deposits, withdrawals, interest calculation, and fund transfer.
9. Display account details before and after transactions.

Steps for Execution

1. Create the Java project and add Account, SavingsAccount, CurrentAccount, and BankSystem.
2. Compile and run BankSystem.java.
3. Observe the initial account details.
4. Perform deposit and withdrawal operations.
5. Calculate savings interest.
6. Perform the current-account overdraft withdrawal.
7. Transfer funds between accounts.
8. Observe the final balances.

Source Code

```
class Account {
    int accountNumber;
    String accountHolderName;
    double balance;
    String accountType;

    Account(int accountNumber, String accountHolderName,
            double balance, String accountType) {
        this.accountNumber = accountNumber;
        this.accountHolderName = accountHolderName;
        this.balance = balance;
        this.accountType = accountType;
    }

    void deposit(double amount) {
        balance = balance + amount;
        System.out.println("Deposited: " + amount);
    }
}
```

```

void withdraw(double amount) {
    if (amount <= balance) {
        balance = balance - amount;
        System.out.println("Withdrawn: " + amount);
    } else {
        System.out.println("Insufficient balance.");
    }
}

void transfer(Account anotherAccount, double amount) {
    if (amount <= balance) {
        balance = balance - amount;
        anotherAccount.balance = anotherAccount.balance + amount;
        System.out.println("Transferred: " + amount);
    } else {
        System.out.println("Insufficient balance for transfer.");
    }
}

void displayAccountDetails() {
    System.out.println("\n--- Account Details ---");
    System.out.println("Account Number : " + accountNumber);
    System.out.println("Holder Name      : " + accountHolderName);
    System.out.println("Account Type    : " + accountType);
    System.out.println("Balance        : " + balance);
}
}

class SavingsAccount extends Account {
    double interestRate;

    SavingsAccount(int accountNumber, String accountHolderName,
        double balance, double interestRate) {
        super(accountNumber, accountHolderName, balance, "Savings");
        this.interestRate = interestRate;
    }

    void calculateInterest() {
        double interest = balance * interestRate / 100;
        balance = balance + interest;
        System.out.println("Interest Added: " + interest);
    }
}

class CurrentAccount extends Account {
    double overdraftLimit;

    CurrentAccount(int accountNumber, String accountHolderName,
        double balance, double overdraftLimit) {
        super(accountNumber, accountHolderName, balance, "Current");
        this.overdraftLimit = overdraftLimit;
    }

    @Override
    void withdraw(double amount) {
        if (amount <= balance + overdraftLimit) {
            balance = balance - amount;
            System.out.println("Withdrawn: " + amount);
        } else {
            System.out.println("Withdrawal exceeds overdraft limit.");
        }
    }
}

public class BankSystem {
    public static void main(String[] args) {
        SavingsAccount savings =
            new SavingsAccount(101, "Ravi", 10000, 5);

        CurrentAccount current =
            new CurrentAccount(102, "Arun", 5000, 3000);

        System.out.println("BEFORE TRANSACTIONS");
        savings.displayAccountDetails();
        current.displayAccountDetails();
    }
}

```

```
System.out.println("\n--- Deposit ---");
savings.deposit(2000);

System.out.println("\n--- Savings Withdrawal ---");
savings.withdraw(3000);

System.out.println("\n--- Interest Calculation ---");
savings.calculateInterest();

System.out.println("\n--- Current Withdrawal ---");
current.withdraw(7000);

System.out.println("\n--- Fund Transfer ---");
savings.transfer(current, 2000);

System.out.println("\nAFTER TRANSACTIONS");
savings.displayAccountDetails();
current.displayAccountDetails();
    }
}
```